

 Papers RAG V2.5

Semantic Search & Deep Chat over your PDF library

Papers Folder

Copy the full path to your papers folder.

Papers root 

/home/mneira/MAURICIO/papers

Index root: papers-rag_index/

Synchronize Databases

Recommended after adding, deleting, or renaming PDFs. Runs the update chain in order and writes TXT and HTML health reports.

 Synchronize/Build/Update Databases

1. Full-Text PDF Vector Database

 PDF vector database ready
839 indexed PDFs · 101,086 chunks

Built from parsed PDF chunks. Used for body-text semantic search. Current PDF folder: 845 PDFs. Model: BAAI/bge-small-en-v1.5

PDF folder and vector database need attention: 6 PDFs present on disk but not currently indexed · 0 indexed PDFs missing from disk. Run Synchronize Databases, or use this update button followed by the keyword index rebuild.

 Build/Update PDF Vector Database

 Semantic Search  Deep Chat with Gemini

 Semantic Search

Paste PDF basenames

One filename per line (no paths required), case-insensitive. Resolves against the indexed corpus. Works without running a search.

paper_one.pdf
paper_two.pdf

Apply pasted names

Clear manual-add list

Clear all (search, paste, selections)

Boolean search (Clause rows · groups · AND / OR / NOT)

Up to 8 rows (Clause 1, Clause 2, ...). Optional groups: choose where a new group starts after a Clause; operators on those edges combine whole groups. Each row is semantic or keyword; keyword rows use token/phrase search and support + prefix matches such as neuro+. Keyword Clause rows bypass the similarity cutoff below.

Add clause

Remove last clause

Start a new group after Clause...



After Clause 2 ×



Clause 1

single-cell RNA

semantic



Within group: Clause 1 with Clause 2

OR



Clause 2

single-cell ATAC

semantic



— Between groups —

Between groups: combine through Clause 2 / Clause 3

AND



Clause 3

guide

semantic



Within group: Clause 3 with Clause 4

OR



Clause 4

Tutor*

keyword

Boolean query

(Clause 1 OR Clause 2) AND (Clause 3 OR Clause 4)

Translation — (semantic) / (keyword) · truncated text · boolean ops are colored: AND · OR · NOT.

(semantic) single-cell RNA OR (semantic) single-cell ATAC

AND

(semantic) guide OR (keyword) Tutor*

Min similarity

0.65

Max results

∞

Semantic discovery source

Union: full-text chunks OR .json metadata

Search

Search wall-clock time: 172.008s

Search diagnostics

No duplicate clauses reused in the last Boolean run.

Session clause cache: 4 / 32 entries

Discovery tuning: PAPER_DISCOVERY_BATCH_CHUNKS=5000 · PAPER_DISCOVERY_MAX_CHUNKS=20000 · EVIDENCE_CHUNKS_PER_PAPER=3

Clear search cache

Paper discovery: 170.302s

Boolean set combination: 0.001s

Evidence retrieval for selected papers: 1.705s; 419 evidence chunks

Returned discovery + evidence chunks: 685

Some semantic clauses reached the discovery chunk ceiling; see the clause-specific notes below.

• Clause 1 (semantic) single-cell RNA — 8.106s; 646 full-text papers · 500 metadata papers · 659 union papers · full-text chunk discovery: 20,000 full-text chunks passed cutoff ≥ 0.65 · 646 representative full-text chunks · considered top 20,000 ranked Chroma chunks · automatic top-N expansion: 5,000 → 10,000 → 15,000 → 20,000 · ceiling reached · more chunks may pass the cutoff beyond this ceiling

Clause 1 note: Automatically raised the Chroma request size: 5,000 → 10,000 → 15,000 → 20,000.

Clause 1 note: Reached the configured discovery ceiling; more chunks may pass the cutoff. Increase PAPER_DISCOVERY_MAX_CHUNKS to search deeper at the same cutoff. Lower the minimum similarity cutoff only if you want broader, lower-similarity matches.

• Clause 2 (semantic) single-cell ATAC — 129.380s; 668 full-text papers · 492 metadata papers · 687 union papers · full-text chunk discovery: 20,000 full-text chunks passed cutoff ≥ 0.65 · 668 representative full-text

chunks · considered top 20,000 ranked Chroma chunks · automatic top-N expansion: 5,000 → 10,000 → 15,000 → 20,000 · ceiling reached · more chunks may pass the cutoff beyond this ceiling

Clause 2 note: Automatically raised the Chroma request size: 5,000 → 10,000 → 15,000 → 20,000.

Clause 2 note: Reached the configured discovery ceiling; more chunks may pass the cutoff. Increase PAPER_DISCOVERY_MAX_CHUNKS to search deeper at the same cutoff. Lower the minimum similarity cutoff only if you want broader, lower-similarity matches.

- Clause 3 (semantic) guide — 0.944s; 6 full-text papers · 0 metadata papers · 6 union papers · full-text chunk discovery: 8 full-text chunks passed cutoff ≥ 0.65 · 6 representative full-text chunks · considered top 5,000 ranked Chroma chunks · automatic top-N expansion: 5,000 · cutoff boundary reached; scan stopped after lower-scoring chunks

Clause 3 note: Checked the top 5,000 ranked chunks; no larger request was needed.

Clause 3 note: Stopped because the scan crossed below the similarity cutoff.

- Clause 4 (keyword) Tutor* — 2.187s; 154 representative chunks · 154 papers · FTS-backed keyword discovery

> 📁 Search results — 153 papers · **Boolean query:** (Clause 1 OR Clause 2) AND (Clause 3 OR Clause 4) · *Similarity filter: semantic ≥ 65% (keyword Clause rows exempt).*

Select papers — Quick Chat & export use checked rows; Send to Deep Chat loads all checked PDFs.

All 153 paper(s) selected.

Select all

Deselect all

> 📁 Search hit papers — 153 paper(s)

📄 Export context for external LLM

💬 Send selected papers to Deep Chat →

💬 Quick Chat (Vertex)

Checked papers only: abstract JSON for each; best excerpt when the paper came from search hits. Filename citations required.

Ask about the checked papers... 

 Find the papers that are actually tutorials and deal with both scRNA and scATAC together, how many such papers?

 There are 3 papers that are identified as tutorials or describe workflows dealing with both single-cell RNA sequencing (scRNA-seq) and single-cell ATAC sequencing (scATAC-seq) together:

1. [140_Fabian_Theis_best_practices_single_cell_omics_nature_methods_2023.pdf](#) : This article guides readers through steps of unimodal and multimodal single-cell data analysis, suggesting comprehensive best-practice workflows for various analysis steps and serving as an entry point for novices and a guide for advanced users in single-cell (multi-)omic analysis [Abstract · [140_Fabian_Theis_best_practices_single_cell_omics_nature_methods_2023.pdf](#) , p.2].
2. [382_jointly_defining_cell_types_from_multile_single_cell_datasets_using_LIGER_2020.pdf](#) : This paper provides a step-by-step protocol for using LIGER to jointly define cell types from single-cell RNA-seq and single-nucleus ATAC-seq data. The protocol covers data preprocessing, normalization, joint factorization, quantile normalization, joint clustering, and visualization [Abstract · [382_jointly_defining_cell_types_from_multile_single_cell_datasets_using_LIGER_2020.pdf](#)].
3. [306_panpipes_pipeline_for_multiomic_and_spatial_transcriptomic_data_analysis_Fabian_Theis_2024.pdf](#) : This paper presents Panpipes, a set of computational workflows designed to automate multimodal single-cell and spatial transcriptomic analyses, integrating Python-based tools for quality control, preprocessing, integration, clustering, and reference mapping. It mentions scRNA-seq and scATAC-seq as key molecular modalities [Abstract · [306_panpipes_pipeline_for_multiomic_and_spatial_transcriptomic_data_analysis_Fabian_Theis_2024.pdf](#)].